Environmental Science and Pollution Research (2022) 29:7936–7951
https://doi.org/10.1007/s11356-021-16032-5

RESEARCH ARTICLE

Coupling isotopic analysis and Ecopath model to detect the ecosystem features based on functional groups of the southwestern Yellow Sea, China

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Received: 30 September 2020 / Accepted: 14 August 2021 / Published online: 4 September 2021

Abstract

The paper evaluates the species richness, material transfer, energy flow, and system function of the southwestern Yellow Sea Ecosystem (SYSE) indicating intensive human intervention affecting this large marine ecosystem. Twenty functional groups were chosen to represent the basic components of the SYSE for Ecopath modeling based on offshore surveys, annual bird observations, and the China Fisheries Statistical Yearbooks. Forty-nine species based on 15 functional groups of Ecopath model were assessed by stable isotopic analysis (SIA) to verify ecosystem features, energy flow, and trophic structure of the SYSE derived from Ecopath model. Results showed there was a clear correlation of the estimated trophic structure calculated from SIA and the Ecopath model with R^2=0.7184. The SYSE Ecopath model was still at an immature and unstable stage according to outputs of the modeling parameters. This paper provides a verification method of detecting the ecosystem features and maturity, stability, and resilience of marine ecosystems by comparing outputs from Ecopath models with SIA.

Keywords Ecopath model · Trophic structure · Functional groups · Stable isotope analysis (SIA) · southwestern Yellow Sea Ecosystem (SYSE) · China

Introduction

Understanding the structure and function of aquatic ecosystems is fundamental to the wise management of all waterbodies on the earth's surface. Scientists have developed many quantitative approaches to measure the energy flow of various ecosystems, such as the habitat suitability model (Ottaviani et al. 2004), the maximum entropy model (Phillips et al. 2006), the Ecopath with Ecosim model (Pauly et al. 2000; Villasante 2016), and the cross-wavelet model (Shuai et al. 2018). The Ecopath model has been widely used and is considered one of the core tools for aquatic ecosystem study among many models since its first usage (Polovina 1984) and theoretical module completion (Ulanowicz and John 1988). It is much more popular with scientific interpretation at present with the development of computer application software (Christensen and Walters 2004) and expansion to 3D Ecopath, Ecosim, and Ecospace modules (Walters et al. 2005). Its study sites cover a range of aquatic ecosystems, from lakes (Bacalso et al. 2016; Heymans and Tomczak 2016), estuaries (Carrer et al. 2000; Bacalso and Wolff 2014; Mcgill et al. 2017), and bays (Morales-Zárate et al. 2004; Mutsert et al. 2016) to offshore areas (Pauly et al. 1998, 2000; Harvey et al. 2003; López et al. 2008; Bayle-Sempere et al. 2013; Tecchio et al. 2013; Whitehouse et al. 2014; Reinaldo et al. 2016; Coll et al. 2016) in both terrestrial and marine waterbodies. The abovementioned studies have provided a clear sketch of the structure, function, and status of many aquatic ecosystems at the global level.

Studies have shown that biological metabolism causes fractional distillation of isotopes, such as respiration (carbon) and excretion (nitrogen), which preferentially emits lighter isotopes and retains heavier isotopes to varying degrees (Deniro and Epstein 1978). Among them, the carbon stable isotope composition of consumers is much more similar to its food source than to the nitrogen stable isotope composition. It is

Responsible Editor: V.V.S.S. Sarma

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common to determine the trophic level of an organism in a food web by identifying 3~5‰ increases of the nitrogen stable isotope ratio comparison between its consumers and predators (Hobson and Welch 1992). Therefore, the stable isotope composition in the organism can reflect the characteristics of the food ingested by the organism over a relatively long period of life (Hesslein et al. 1993). Those data can reflect the nutrient source of the initial producers, the food sources and composition of the consumers, the trophic level of different species/organisms, the structure of the food webs, and the energy flow pathways in various ecosystems. Stable isotopic analysis (SIA) has been widely used to determine the food composition of organisms, tracking food sources and nutrient dynamic processes in many fields (Drazen et al. 2008; Fanelli et al. 2011).

The Yellow Sea, the third largest marginal sea in China and one of the 64 large marine ecosystems in the world, lies in the western part of the Pacific Ocean facing the Korean Peninsula (Fig. 1). The Yellow Sea Large Marine Ecosystem (YSLME) is a highly productive sea area but harbors on of the largest coastal populations in the world, and is therefore possibly one of the most human-impacted large marine ecosystems worldwide (Walton 2010; Kearney et al. 2013). The Yellow Sea LME's ecosystem carrying capacity (ECC) provides more than 2×10^6 t/a in capture fishery and 1.4×10^9 t/a in mariculture, supporting wildlife, beaches, and spaces for tourism. There are over 339 fishes, 100 polychaetes, 171 mollusks, 107 crustaceans, and 22 echinoderms recorded in YSLME. At least two million shorebirds use the region as a stopover during migration, representing approximately 40% of all migratory shorebirds using the East Asian-Australasian Flyway (UNDP/GEP Yellow Sea Large Marine Ecosystem 2019). Five large coastal cities comprised of a population in the tens of millions border the sea: Qingdao, Dalian, and Shanghai in People's Republic in China; Seoul/Incheon in Republic of Korea; and Pyongyang/Nampo in the Democratic People's Republic of Korea (D.P.R. Korea). The natural resources are both vulnerable and limited, threatened by the intensive anthropogenic activities in the YSLME. However, resource demands will continue to impose serious pressure on ecosystems and our global footprint in the near future as human population increases, accompanied by rapid economic development in the surrounding coastal area. Figure 1 shows the sampling sites of our comprehensive survey in SWYE.

In recent years, many studies focused on one or more key species, dominant species and species composition (Dong et al. 2004; Zhang et al. 2009a, 2009b; Zhang 2012; Wu et al. 2015), quantitative change (Nie 2008; Yang et al. 2010; Zhang et al. 2014), and other temporal and spatial variation characteristics (Meng 2011; Wu 2011; Jin et al. 2015; Wang et al. 2015) in the Yellow Sea ecosystem. However, it is necessary to consider complicated coastal and marine ecosystems as entities including detritus, plankton, small and large benthic organisms, nekton, and birds. The Ecopath model can provide us with a reliable method to recognize complicated aquatic ecosystems based on mass-balanced modeling. This paper evaluates the species richness, material transfer, energy flow, and system function of the SYSE based on offshore surveys, the annual bird observations, and the China Fisheries Statistical Yearbooks. It provides a good example of how to evaluate the complicated characteristics of marine ecosystems based on both SIA and the Ecopath model. Furthermore, the results can also serve as useful guidance and instruction for ecosystem restoration and fishery resources management in YSLME.

Methods

Ecopath model

The Ecopath model assumes that an ecosystem consists of a series of functional groups with similar biological and ecological characteristics, ensuring that all functional groups cover the energy flow of the ecosystem. Based on the principle of nutritional balance, the model defines the energy input and output of each functional group in the ecosystem can be balanced, as:

Production - Predation death - Natural death - Output = 0

The model uses a set of linear simultaneous equations (Christensen et al. 2005) to define an ecosystem, each linear equation representing a functional group in the ecosystem:

$$\mathrm{B_i} \cdot \left(\frac{\mathrm{P}}{\mathrm{B}}\right)_{\mathrm{i}} \cdot \mathrm{EE_i} = \sum_{\mathrm{j=1}}^{\mathrm{n}} \mathrm{B_j} \cdot \left(\frac{\mathrm{Q}}{\mathrm{B}}\right)_{\mathrm{j}} \cdot \mathrm{DC_{ji}} + \mathrm{EX_i} \quad (1)$$

where B_i is the biomass of functional group i; P_i is the production of functional group i; $(P/B)_i$ is the ratio of production to biomass of the functional group i; EE_i is the ecological nutrition conversion efficiency of the functional group i; $(Q/B)_j$ is the digestion ratio of consumption to biomass; DC_{ji} is the food composition matrix; and EX_i is the output of catch and migration.

B_i, $(P/B)_i$, $(Q/B)_j$, EE_i, DC_{ji}, and EX_i are the basic parameters for model inputs. Any unknown number appearing in the first 4 parameters can be calculated by other parameters in the modeling. The latter two parameters DC_{ji} and EX_i are required to input parameters.

Basic data information

Comprehensive ecosystem surveys (2006–2009) results provided the data of phytoplankton, zooplankton, benthic organisms, swimming animals, and environmental factors covering the nearshore waters of the southwestern Yellow Sea. All sampling and surveys were completed according to the

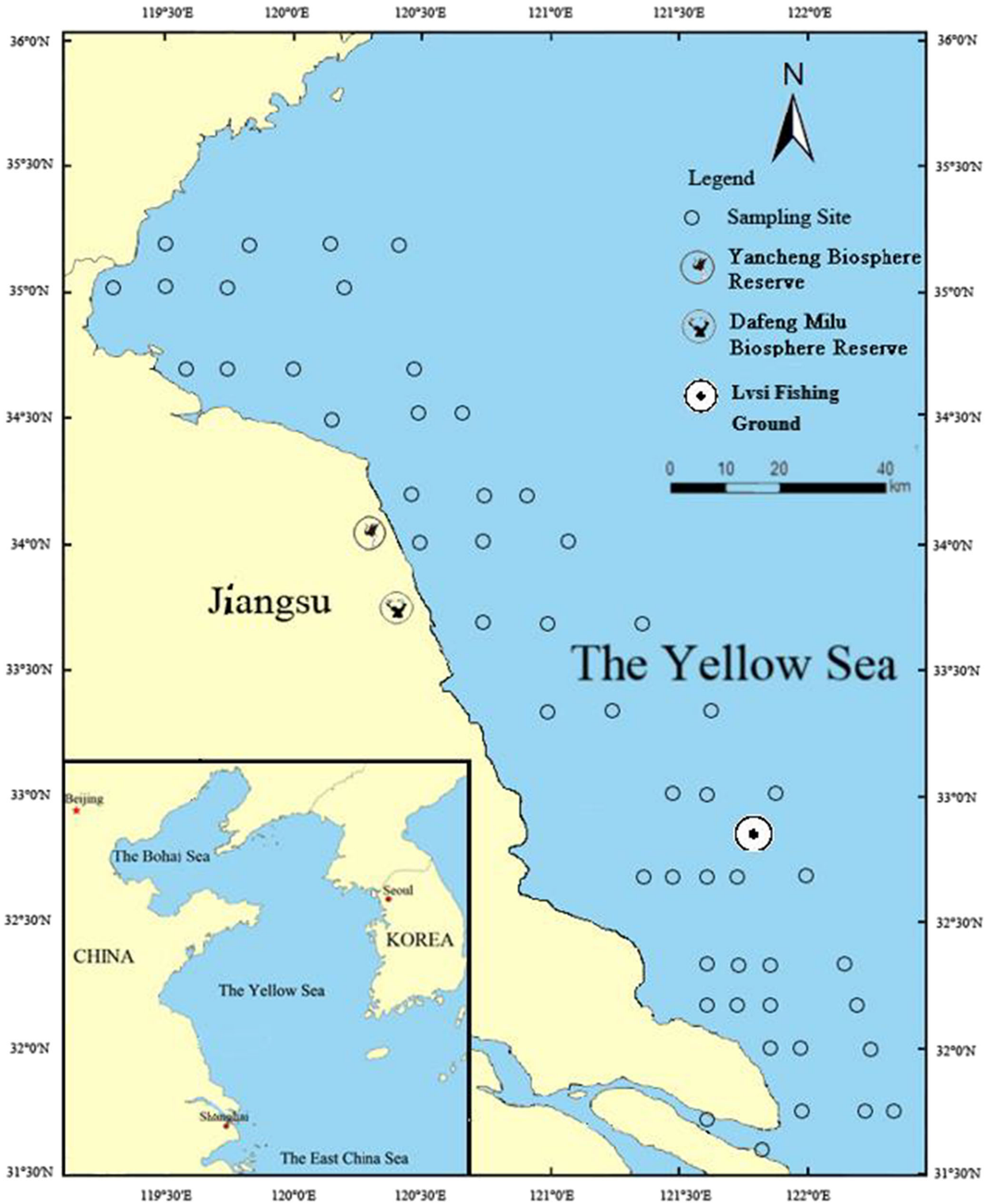


Fig. 1 Sample sites and study area in the southwestern Yellow Sea

technical guidelines meeting the National Standards of Marine Monitoring (GB17378-2007), Marine Survey Specification: Marine Biological Survey (GB12763.6-2007), and Environmental Monitoring of Coastal Waters (HJ442-2008).

Forty-nine species from 15 functional groups were collected (2018) and grouped with the help of local fishermen for stable isotopic analysis. Multi-year bird survey data were provided with the courtesy of UNESCO Yancheng World Biosphere Reserve. Fishery data referred to the China Fisheries Statistical Yearbooks which were published by the Ministry of Agriculture, China. There were only two comprehensive surveys carried out during 1985–1988 and 2006–2008 in national level in this area, without ongoing surveys in recent years (Wang et al. 2019). That is why there is no new data for modeling though the paper prefers to use data from current survey. The local marine monitoring station around this area only focuses on their annual monitoring plan and cannot provide the data covering all functional groups needed for Ecopath model. Therefore, we assumed there were no significant changes happened in the southwestern Yellow Sea ecosystem since 2008.

Sources of ecological parameters of 20 functional groups

The on-site marine bio-ecological survey provided the functional group biomass (B) for the SYSE Ecopath model. The fishery resource biomass data were calculated by the swept area method (Ungaro et al. 1996). The natural mortality coefficient was estimated using the empirical formula developed by Pauly (1980). The total catch curve method (Gulland 1971) helped to compute the instantaneous total mortality value Z. The fishery production/biomass (P/B) under the ecosystem equilibrium condition was equal to Z. The consumption/biomass (Q/B) values of fish were computed according to the multiple regression model of the caudal fin shape ratio proposed by Palomares and Pauly (1998). It was difficult to determine the P/B and Q/B values of detritus, phytoplankton, and zooplankton based on our survey. Thus, the paper referred to the similar functional groups, combined with data from several online databases (www.fishbase.org, www.sealife.org, www.seaaroundus.org) to determine the P/B and Q/B parameters for modeling. The diet composition (DC) matrix of functional groups was acquired from the gastric contents and historical data of sampled fish (Cui 1985; Yang 2001a, 2001b, 2001c; Song et al. 2014; He et al. 2012). Ecotrophic efficiency (EE) indicates the conversion efficiency of the contribution of each functional group to the energy contribution of the ecosystem. The EE value ranges from 0 to 1, which is difficult to measure directly. Usually, it is set as an unknown parameter and the Ecopath model is debugged. Its parameter acquisition was achieved by bringing the system to equilibrium.

System parameter indicators

The trophic level of each functional group used the concept of fractional trophic level proposed by Odum (1971). The trophic level of each organism was based on the trophic level of its food intake (defining the nutrient levels of detritus and phytoplankton to 1) and its proportion in the food composition was weighted. The concept of trophic level, also known as effective trophic level (ETL), could better express the nutritional structure of functional groups in ecosystems. In order to facilitate the analysis of energy flow and system energy distribution between various nutrient levels, a method of trophic level polymerization of the system was used to simplify the complicated food network. Therefore, the energy flows from different functional groups combined into several trophic levels could be expressed as an integer.

Many indicators in the Ecopath model were used to describe important characteristics such as system size, stability, and maturity. The ratio of total primary production/total respiration (TPP/TR) was the key indicator of ecosystem maturity. The further the value was from 1, the more residual production in the system could be reused. Thus, the ecosystem utilization rate was inefficient with low maturity. The connectance index (CI) and the system omnivory index (SOI) were indicators that characterized the complexity of functional groups within an ecosystem. SOI described the degree of predation allocation among functional groups at different trophic levels. CI referred to the ratio of the number of connections actually existing in the food web to the number of possible connections. Finn's cycling index (FCI) was the ratio of system recirculation flow to total flow, which characterized the flow rate of ecosystem organisms. Finn's mean path length (MPL) referred to the average length of the food chain. Both FCI and MPL characterized the maturity of the ecosystem (Finn 1976).

Stable isotope analysis

Sampling and pretreatment

All sampling was pretreated under the direction of the Marine Survey Specification: Marine Biological Survey (GB12763.6-2007). We collected, classified, and selected biological samples in situ, marked with specimen bags and labels, and then kept them at 0°C in a portable fridge. Basic characteristics such as body length, size, and weight were measured in our laboratory. We used scalpels, forceps, scissors, and other tools to extract muscles from samples, such as the back muscles of fishes, abdominal muscles of shrimps, first chela muscles of crabs, adductor muscles from shellfish, and carcasses from cephalopods. We used a freeze dryer (ALPHA-1-4, Martin Christ, Germany) to dry samples at −40 ° C to a constant weight, and then ground them to powder passing 200 mesh sieve with agate mortar. Finally, we stored them in desiccators for carbon and nitrogen stable isotope analysis.

Stable isotope measurement

All pretreated samples were packed into tin boats for compaction. The carbon and nitrogen stable isotope composition was analyzed by an isotope ratio mass spectrometer (Delta V Advantage, Thermo Fisher Scientific, USA). Function (2) showed the stable isotopic composition of carbon and nitrogen by the value of δ:

$$\delta_{\text{sample}}(\text{‰}) = \left(\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1\right) \times 1000 \quad (2)$$

where δ_{sample} represents the $\delta^{13}C$ or $\delta^{15}N$ value of the sample, R_{sample} represents the $^{13}C/^{12}C$ or $^{15}N/^{14}N$ ratio of the measured sample, and $R_{standard}$ represents the $^{13}C/^{12}C$ or $^{15}N/^{14}N$ ratio of the reference material. The reference standards for carbon and nitrogen isotope composition are the pseudo-stone standard material (PDB) of the Cretaceous Pidi group in South Carolina, USA, and atmospheric nitrogen. In order to verify the stability of the instrument and the accuracy of the measured values, during the carbon stable isotope analysis, one part of the IAEA-CH-3 standard material ($\delta^{13}C$ = −24.7‰) was inserted into each of 10 samples for drift analysis. During the nitrogen stable isotope analysis, one part of the IAEA-NO_3 standard material ($\delta^{15}N$ = 4.7‰) was inserted for ten samples per interval for drift analysis. The relative standard deviation of the measured $\delta^{13}C$ and $\delta^{15}N$ was less than 0.2‰. Stable carbon and nitrogen isotopic composition were measured at the laboratory of Nanjing Institute of Geography and Limnology, Chinese Academy of Sciences.

Consumer enrichment of ^{15}N is more obvious than ^{13}C in the food chain transfer, so the nitrogen isotope composition is used to evaluate the consumer's trophic level in the food web, which is calculated as follows (Post 2002):

$$\text{Trophic level} = \lambda + \frac{\delta^{15}N_{\text{consumer}} - \delta^{15}N_{\text{base}}}{\Delta\delta^{15}N} \quad (3)$$

where $\delta^{15}N_{base}$ is the nitrogen stable isotope ratio of the primary producer or primary consumer of the ecosystem food web. When λ = 2, $\delta^{15}N_{base}$ is the $\delta^{15}N$ value of the primary consumer. In this study, the average value of $\delta^{15}N$ of the bivalve mollusk was used as the isotope baseline value, $\delta^{15}N_{consumer}$ was the stable nitrogen isotope ratio of the consumer, and $\Delta\delta^{15}N$ was the enrichment value during the trophic level transfer (average value was taken as 3‰).

Cluster analysis

Based on the squared Euclidean distance of stable carbon and nitrogen isotope values from sampling organisms, we used the between-groups linkage cluster analysis of IBM Statistical Product and Service Solutions Statistics 25 (SPSS 25) to divide the nutrition groups of the main species of food web in the southwest Yellow Sea. The cluster analysis icicle and the cluster analysis dendrogram can intuitively display the results of cluster analysis from different angles. If the two functional groups are not separated by color bars in the cluster analysis icicle, it means that the two functional groups have been merged into one category at this stage; otherwise, they are divided into two different categories. The whole process of clustering and merging could be clearly analyzed from the cluster analysis icicle, and the cluster analysis dendrogram illustrates the merging of functional groups step by step. The final classification results should be determined by comprehensive consideration and selection criteria.

Results

Functional groups and main species in SYSE

Twenty functional groups were selected to construct the Ecopath model based on biological survey data according to the biological species, feeding, and biological characteristics of the SYSE. Functional groups and their key species covered nearly the entire structure and energy flow process of the SYSE (Table 1).

Nutritional structure and energy flow distribution

The Ecopath model defined the trophic level of phytoplankton and detritus in SYSE as 1. Table 2 shows the nutritional structure in SWSE. The models found that the trophic level of the zooplankton and benthic meiobenthos was approximately 2 while the trophic level of the main fish species targeted by fisheries was in the range of 3.154~4.509. The trophic level of the Southwest Yellow Sea ecosystem ranged from 1 to 4.509, of which Dasyatidae was the highest, followed by Anguilliformes.

The SYSE was composed of seven integrated nutrient levels by trophic level aggregation (Table 3). The food intake, output, and total flow in nutrient grades VII and VI were low, and the overall energy flow distribution was a typical pyramid. The food intakes of trophic level I and II were 896.4 and 100.5 t/km^2/a, respectively, accounting for 88.75% and 9.95% in total. The energy flow in trophic grades I and II was 2966 and 896.4 t/km^2/a, respectively, accounting for 74.21% and 22.43% of the total system, respectively. Both occupied 96.64% of the whole ecosystem. While trophic levels IV, V, and VI only occupied 0.29%, 0.035%, and 0.004% of the total system flow, respectively. The VII trophic level was almost negligible indicating the lowest trophic energy flow. It showed that the energy flow of the low trophic level accounted for a large proportion in the system, and the low trophic level composed of primary producers (phytoplankton) was the main source of system energy. From nutrient flow to

Table 1 Functional groups and main species in the SYSE

No.	Functional groups	Group description
1	Charadriiformes	*Calidris alpina*, *C. ruficollis*, *Charadrius alexandrinus*, etc.
2	Anseriformes	*Anser*, *Anser bernicla*, *Anas platyrhynchos*, *Anas poecilorhyncha*, *Tadorna*, *Mergus squamatus*, etc.
3	Gruiformes	*Grus japonensis Grus grus*, *Grus leucogeranus*, *Grus monacha*, etc.
4	Other benthic fishes	Scorpaeniformes, Tetraodontiformes, Gobies, Syngnathidae, Lophiidae, Leiognathidae, Sparidae, Sillaginidae, Ariilae, Pomadasyidae, Chaetodontidae, Mugiloididae, Pholidae, Zoarcidae, Callionymidae, etc.
5	Other pelagic fishes	Clupeidae, Trichiuridae, Synodontidae, Scombridae, Scopelidae, Mugilidae, Polynemidae, Sphyraenidae, Salangidae, Chimaera, Carangidae, Ammodytidae, etc.
6	Dasyatidae	*Dasyatis akajei.*
7	Anguilliformes	Congeridae, Muraenesocidae, *Ophichthus apicalis*
8	Pleuronectiformes	*Cynoglossus (Areliscus) semilaevis*, *Cynoglossus (Areliscus) abbreviatus*, *Pleuronichthys cornutu*, Bothidae, Soleidae, etc.
9	Sciaenidae	*Miichthys miiuy*, *Larimichthys polyactis*, *Collichthys lucidus*, *Nibea albiflora*, *Johnius belengeri*, etc.
10	Engraulidae	*Setipinna taty*, *Thryssa kammalensis*, *Coilia mystus*, *Thryssa mystax*, etc.
11	Stromateidae	*Pampus argenteus*, *Psenopsis anomala*, etc.
12	Decapod	*Oratosquilla oratoria*, *Metapenaeopsis dalei*, *Palaemon gravieri*, etc.
13	Brachyura	*Portunus trituberculatus*, *Matuta planipes*, *Dorippe japonica*, *Charybdis japonica*, etc.
14	Cephalopoda	*Octopus ocellutus*, *Octopus ocellutus*, *Loligo japonica*, *Sepiola birostrata*, *Sepia esculenta*, *Sepiella maindroni*, *Todarodes pacificus*, etc.
15	Benthic macrobenthos	Body length > 1 mm, including polychaetes, Crustacean, Echinoderms, mussels and oysters, etc.
16	Benthic meiobenthos	Body length < 1 mm, including nematodes, copepods, ostracods, polychaetes, etc.
17	Jellyfishes	Hydrozoa, Scyphozoa
18	Zooplankton	Chaetognatha, mysids, fish eggs, larvae, ostracods crustaceans, bivalve larvae, barnacle larvae, Cladocera, echinoderms larvae, etc.
19	Phytoplankton	Diatoms, dinoflagellates, cyanobacteria, and other phytoplankton
20	Detritus	Particulate organic carbon and dissolved organic carbon

detritus, the total amount of trophic levels I and II to detritus was 1081 t/km^2/a and 216.7 t/km^2/a, separately accounting for the total amount of detritus of 81.22% and 16.28%, indicating that nearly all energy flow in the system was utilized for trophic level II. The ecosystem also made full use of the medium and high trophic fractions of trophic level III and above. Due to the energy utilization being insufficient in trophic level I, the energy transfer might be blocked between trophic level I and II.

Stable isotope analysis

Our samples for stable isotope analysis covered 15 functional groups in SYSE. Table 4 and Figure 2 showed their δ^{13}C and δ^{15}N values within the 95% confidence interval. The 15 functional groups are as follows: Charadriiformes, Anseriformes, Gruiformes, other benthic fishes, other pelagic fishes, Dasyatidae, Anguilliformes, Pleuronectiformes, Sciaenidae, Engraulidae, Decapod, Brachyura, Cephalopoda, benthic macrobenthos, and jellyfish.

Compared with the Ecopath model results, SIA showed most functional groups were equally located in the overall nutritional structure of the ecosystem. The distribution of δ^{13}C and δ^{15}N values of macrobenthos and other benthic fishes was relatively scattered with the large variety of samples. The δ^{15}N values of Anseriformes and Gruiformes were lower than results from the Ecopath model as omnivores consume more seeds and land-based plants. In addition to the animals contained in the model functional group, plant seeds and fruits also accounted for a large proportion of their diet composition. However, foods derived from land were not included in the Ecopath model. The δ^{15}N value of some adult fishes, such as Engraulidae, was higher than that in its juvenile stage. The δ^{15}N values of jellyfish were lower because only *Rhopilema esculenta* Kishinouye and *Rhopilema*

Table 2 Input and output parameters of Ecopath model in SYSE

Functional groups	Trophic level	B (t/km^2/a)	P/B	Q/B	EE
Charadriiformes	3.741	0.100	0.10	4.33	0.000
Anseriformes	3.779	0.156	0.10	9.28	0.000
Gruiformes	3.760	0.004	0.10	13.00	0.000
Other benthic fishes	3.579	0.147	0.85	2.84	0.999
Other pelagic fishes	3.184	0.096	0.97	5.93	0.963
Dasyatidae	4.509	0.056	1.28	4.65	0.767
Anguilliformes	4.321	0.072	0.98	6.27	0.993
Pleuronectiformes	3.631	0.581	0.97	4.22	0.740
Sciaenidae	3.829	0.078	0.88	4.11	0.914
Engraulidae	3.154	0.183	1.04	7.79	0.998
Stromateidae	2.751	0.229	0.95	4.86	0.956
Decapod	2.901	0.330	7.60	28.90	0.954
Brachyura	3.074	1.140	3.50	12.00	0.666
Cephalopoda	3.417	0.136	3.10	9.30	0.968
Benthic macrobenthos	2.589	9.932	2.00	8.60	0.821
Benthic meiobenthos	2.051	4.723	9.00	33.00	0.354
Jellyfishes	2.968	2.400	5.00	20.00	0.544
Zooplankton	2.020	3.935	25.00	180.00	0.947
Phytoplankton	1.000	8.177	200.00	0.00	0.339
Detritus	1.000	18.000			0.257

esculentum Kishinouye were sampled. Other carnivorous jellyfish at higher trophic levels were not available.

Fifteen major functional groups classification based on SIA

Figure 3 shows the cluster analysis icicle of $\delta^{13}C$ and $\delta^{15}N$ values of major group in SYSE. As shown in Fig. 3, groups 9 and 13 were combined into one category in the first stage. The occurrence of both items in the consolidation came as first place, so the first occurrence of a column was 0. The sequence number of the cancelation of the consolidation result was classified as category 1, and the consolidation result in this stage appeared in stage 5. After 14 stages, all functional groups were combined into one group, and the clustering analysis was finished.

Figure 4 shows the cluster analysis dendrogram of $\delta^{13}C$ and $\delta^{15}N$ values of the major groups in SYSE. Upon studying the different categories in finer detail (clusters 2, 3, 5, and 6), no obvious differences between clusters 2 and 3 were identified since the classification was too coarse. When the number of clusters is 6, there is little difference between the members of 5 and 6 groups. The final number of clusters was 5 after comprehensive consideration. Other pelagic fishes, Pleuronectiformes, Sciaenidae, Cephalopod, and benthic macrobenthos belonged to the first category. The second group included Anseriformes, Gruiformes, and jellyfish. The third group is Engraulidae and Brachyura. The fourth group included

Table 3 Distribution of energy flows at aggregated trophic levels in SYSE (t/km^2/a)

Trophic level	Consumption by predators	Export	Flow to detritus	Respiration	Throughput
VII	0.000327	0.00111	0.0022	0.00553	0.00916
VI	0.00917	0.00993	0.0343	0.0869	0.14
V	0.14	0.068	0.361	0.851	1.42
IV	1.42	0.382	2.99	6.834	11.63
III	11.63	1.608	29.52	57.78	100.5
II	100.5	0.989	216.7	578.2	896.4
I	896.4	988.6	1081	0	2966
Total throughput	1010	992	1331	644	3976

Table 4 Transfer efficiency of discrete trophic levels in SYSE

Source	Trophic level						
	I	II	III	IV	V	VI	VII
Producer	—	11.38	12.75	15.65	14.84	13.43	15.58
Detritus	—	11.24	13.86	15.26	14.36	13.99	15.9
All flows	—	11.33	13.16	15.5	14.67	13.61	15.68
Proportion of total flow originating from detritus: 0.43							
From primary producers: 13.35%							
From detritus: 13.14%							
Total: 13.22%							

Charadriiformes, Dasyatidae, and Anguilliformes. Other benthic fishes occupied the fifth group.

Energy flow between nutrient levels and its conversion efficiency

Figure 5 shows the simplified energy flow process of the SYSE based on a mass-balanced model. Each circle in the figure represents a functional group; the connection between them represents the energy transfer process, and the size of the circle represents the relative biomass. The thickness of the connection represents its proportion in the predator relationship. It also shows that the nutrient flow of the ecosystem in the southwestern Yellow Sea was mainly composed of two pathways: ① a detrital food chain, detritus→ zooplankton and benthic organisms→ small swimming animals→ large fish and birds; ② a food chain, phytoplankton→ zooplankton→ small swimming animals→ large fish and birds.

"Lin's cone analysis" is an effective way to determine the efficiency of energy transfer between integrated nutrient stages. Figure 6 shows that the transfer efficiency between the trophic stages decreased gradually along the food chain. The transfer efficiency from trophic level I was 41.13%, to 22.54% in trophic level II, 2.528% for nutrition level III, and then to nutrition IV with the value of only 0.292%. The consumption and output of living organisms in each trophic level also drastically decreased.

Table 4 shows the energy conversion efficiencies between the various trophic levels derived from primary producers and detritus, respectively. By comparison, the conversion efficiency between high nutrient grades was generally higher than that between low nutrient grades. The conversion efficiency between nutrient grades I→II and II→III was lower, 11.33% and 13.16%, respectively. The highest value of eutrophic conversion efficiency occurred between the vegetative grades VI→ VII; the conversion efficiency from detritus was 15.90%. In general, the conversion efficiency of primary producers with low eutrophication levels was higher than that of source detritus, and the conversion efficiency of primary producers with high trophic levels was lower than that of source detritus. Generally, the total conversion efficiency of the system was 13.22%, which was higher than Lindeman's 1/10 conversion efficiency. The conversion efficiency of source detritus was 13.35%, which was slightly higher than the conversion efficiency of the source primary producers by 13.14%.

Mixed nutrition relationship between functional groups

Predatory or competitive relationships existed directly or indirectly between functional groups within the ecosystem. The interrelationship between functional groups in the Ecopath model is expressed as a mixed nutrient relationship (Fig. 7). The white box indicates that the increase in biomass of the

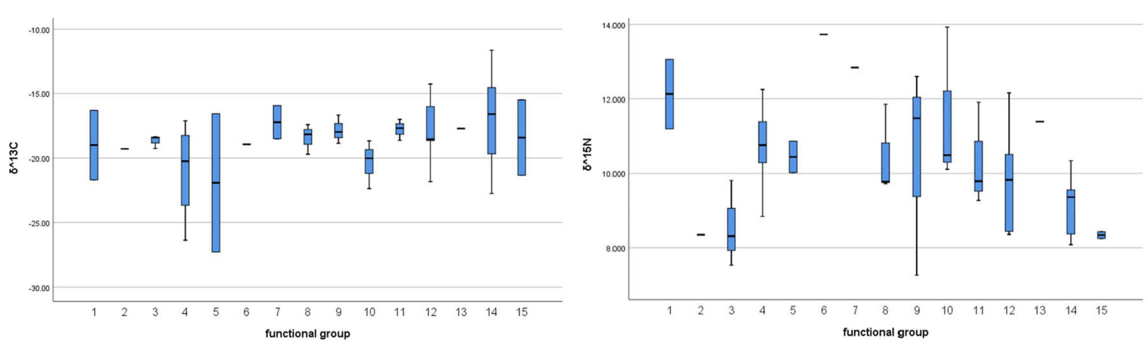
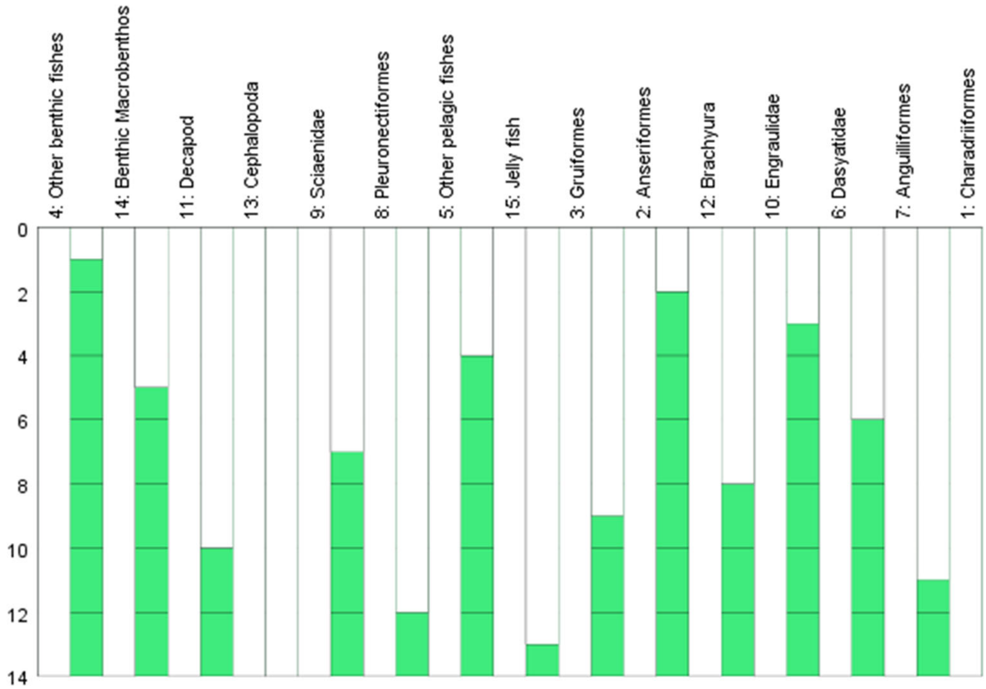


Fig. 2 δ^{13}C (left) and δ^{15}N (right) values box-plots of 15 functional groups in SYSE

Fig. 3 Cluster analysis icicle of $\delta^{13}C$ and $\delta^{15}N$ values of major group in SYSE



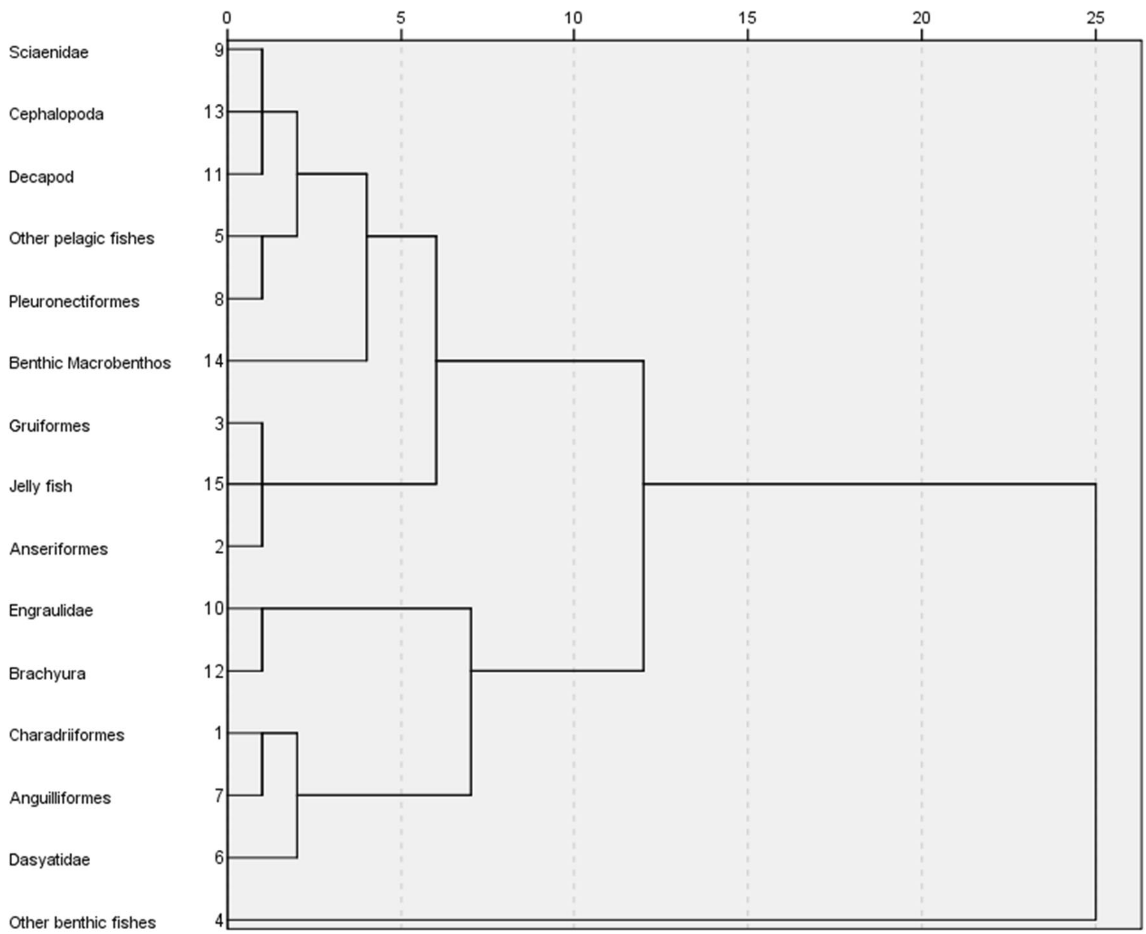


Fig. 4 Cluster analysis dendrogram of $\delta^{13}C$ and $\delta^{15}N$ values of major group in SYSE

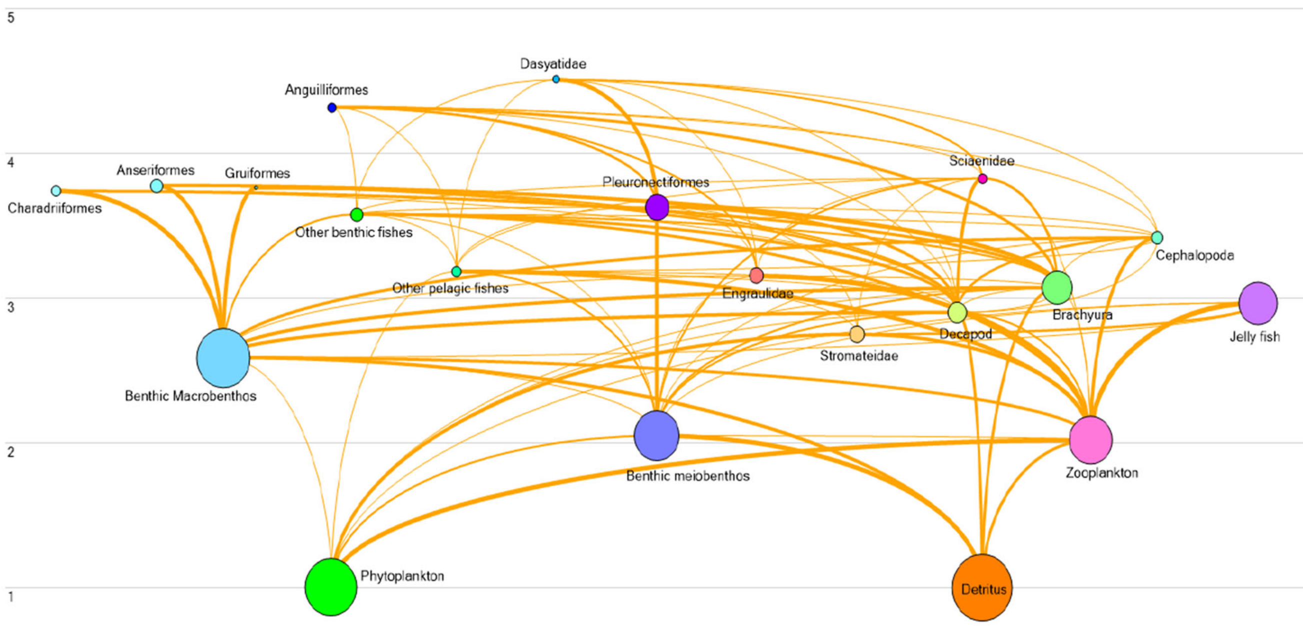


Fig. 5 Energy flow diagram in SYSE Ecopath model

functional group had a positive effect on the increase in biomass of other functional groups, while the black box depicts functional group organisms. The increase in volume had a negative impact on other functional groups, and the height of the box indicates the strength of the impact. It shows that the predator had a positive effect on other functional groups and played a facilitating role, while the predator had a direct or indirect negative impact on other functional groups. Phytoplankton and detritus acted as producers of the system and had a positive impact on most other functional groups. Most functional groups had a direct negative impact on their predators yet had a positive impact on direct predators. The same kind of residual food was negative, and the competition in the same group was particularly severe in the Pleuronectiformes, crab, macrobenthic, jellyfish, and plankton. The negative impact of fishing (fleet) on ecosystems was the most severe one.

Comparison of trophic levels estimated between SIA and the Ecopath model

TLs estimated from the Ecopath model were highly and positively correlated with those estimated from $\delta^{15}N$ values (Fig. 8), which suggests compatible estimations ($y = 0.7926x + 0.5854$, $R^2 = 0.7184$) in TL estimation from both methodologies. Overall, TLs estimated from the Ecopath model were slightly higher than those from SIA for low TL functional groups and lower for high TL functional groups.

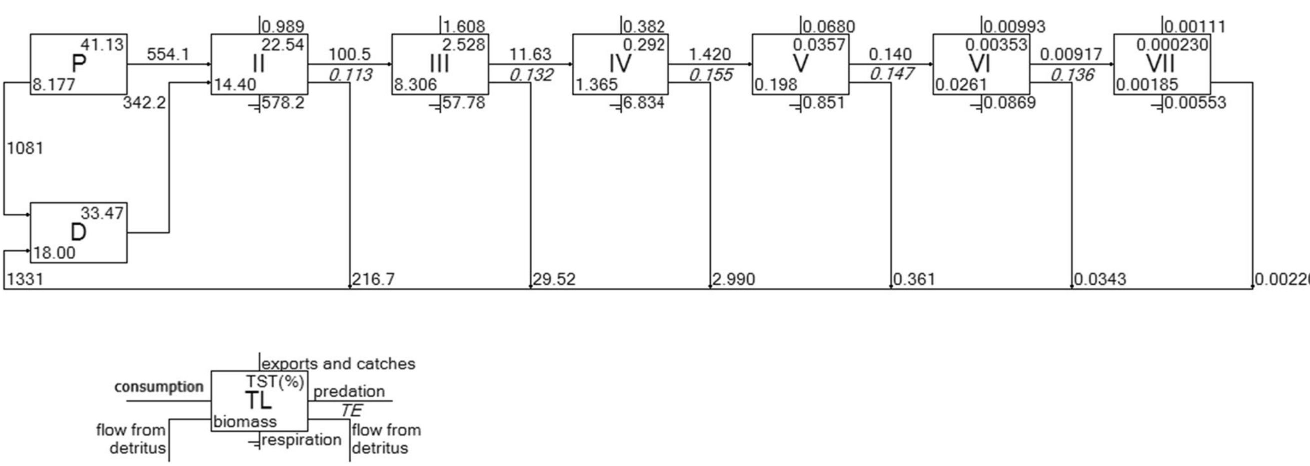


Fig. 6 Trophic flow and transfer efficiency through trophic levels in SYSE

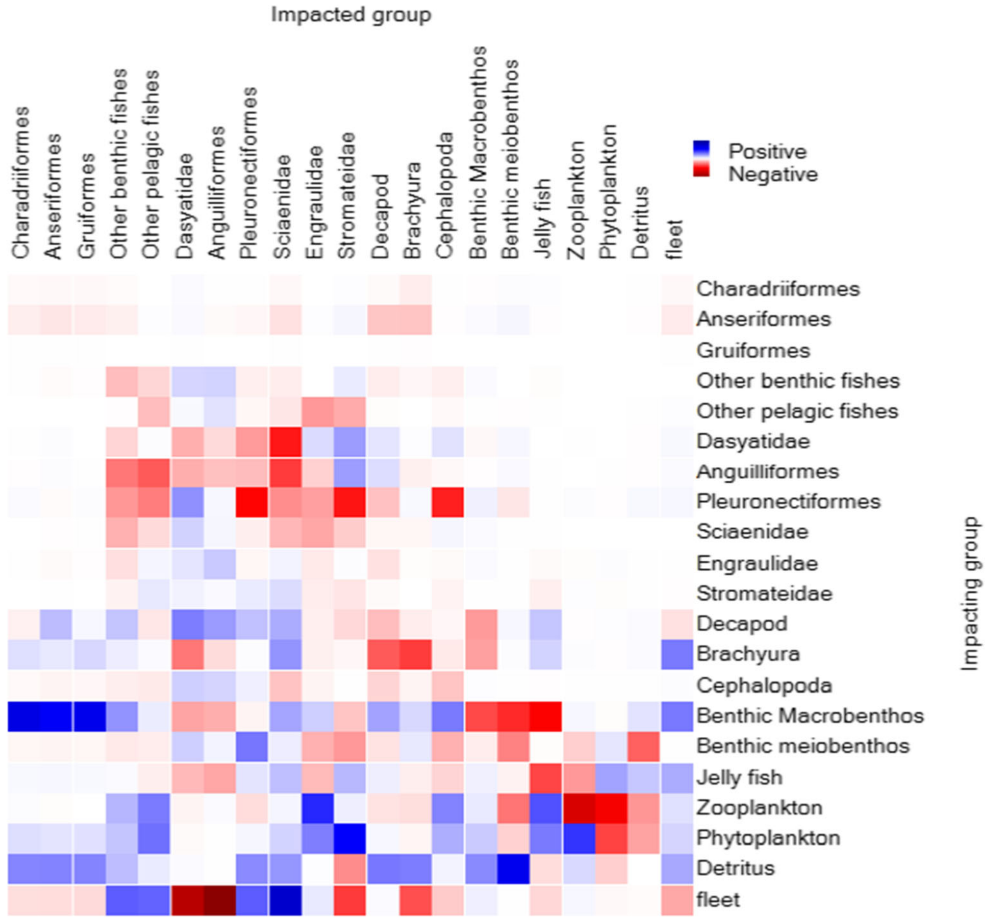


Fig. 7 Mixed trophic impact in SYSE

Generalized characteristics of the SYSE

Highly generalized parameters derived from the Ecopath model can provide a basic profile of the system size, maturity, and stability. Table 5 includes the characteristic parameters of the SYSE. The confidence index of the model system was 0.612. The quality evaluation value of over 150 Ecopath models was between 0.16 and 0.68 (Morissette et al. 2006), demonstrating the high quality and high reliability of the SYSE model. The sum of total respiratory volume, total consumption, total output, and flow detritus in the system was an important indicator to measure the scale of the system. The total flow of the SYSE was 3997.26 t/km^2/a, the total respiration was 643.73 t/km^2/a, and the total consumption was 1031.03 t/km^2/a, accounting for the total

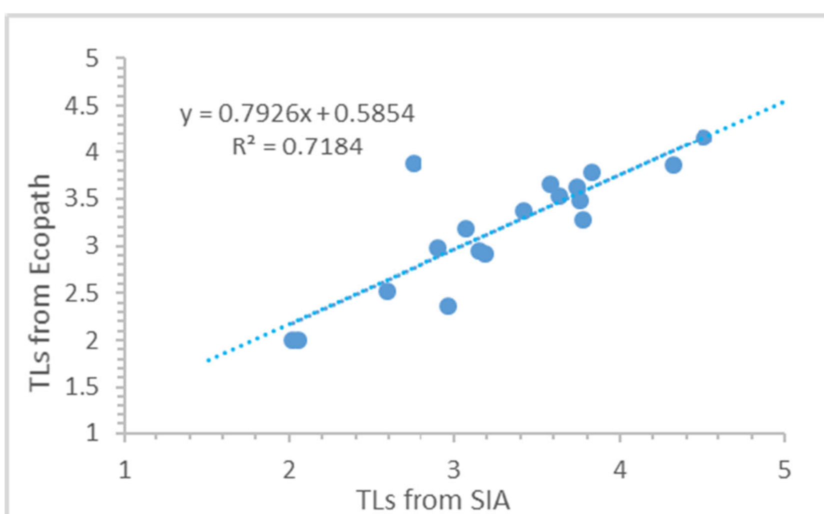


Fig. 8 Linear regression between trophic levels estimated from SIA and Ecopath model for SYSE

Table 5 Generalized statistics in SYSE based on Ecopath model

Parameter	Value
Total system throughput (t/km^2/a)	3997.260
Sum of all consumption (t/km^2/a)	1031.033
Sum of all exports (t/km^2/a)	991.669
Sum of all respiratory flows (t/km^2/a)	643.731
Sum of all flows into detritus (t/km^2/a)	1330.826
Sum of all production (t/km^2/a)	1816.496
Calculated total net primary production (t/km^2/a)	1635.400
Net system production (t/km^2/a)	991.669
Total biomass (excluding detritus) (t/km^2/a)	32.473
Total biomass/total throughput	0.008
Total primary production/total respiration	2.541
Total primary production/total biomass	50.362
Mean trophic level of the catch	2.926
Connectance index	0.280
System omnivory index	0.217
Profit	0.612
Finn's cycling index	3.983
Finn's mean path length	2.444

system flow of 16.10% and 25.79%, respectively. The amount of detritus flow was 1330.83 t/km^2/a, comprising 33.29% of the total system flow. This demonstrates that re-entry into ecosystem recycling was one of the main means of underutilization of system energy flow. In the early stage of ecosystem model development, the production of most functional groups exceeded the amount of respiration with the value of TPP/TR > 1. As the system continued to evolve and the total biomass tended to its maximum, the ecosystem matured and the value of TPP/TB lowered. The TPP/TB of the Southwest Yellow Sea ecosystem was 50.362, which was still in a large range. Total primary production/total respiration was also a key indicator of ecosystem maturity, approaching the value of 1 in a mature ecosystem. The value of the total primary production/total respiration in SYSE was 2.541 reflecting an immature stage. The mean trophic level of the catch in SYSE was 2.926, the system omnivory index was 0.217, and the connectance index was 0.280. The Finn's cycling index and the Finn's mean path length were 3.983 and 2.444, respectively. The above ecosystem parameters showed that the functional groups of the SYSE were closely related, and the structure of the food web was relatively complex, but it was still in a fragile unstable and immature stage.

Discussion

Based on the results of Ecopath model, most of the energy in the SYSE Ecopath model depends on detritus (43%) and primary producers (phytoplankton) (57%), indicating that the grazing food chain dominated the energy flow. The functional group of the SYSE Ecopath model had a trophic level range of 1~4.509. Those functional groups with high trophic levels, such as birds, carnivorous fishes, and cephalopods, occupied the level among 3.417~4.509. Seven integrated trophic levels were recognized based on primary producers. Mixed trophic evaluation showed that phytoplankton and detritus, as producers of the system, had a positive impact on most other functional groups. Interspecific interaction among flounders, crabs, macrobenthos, jellyfishes, and plankton showed negative effects with fierce competition. The overall energy flow distribution showed a typical pyramid type, that is, the value at low nutrient levels was large, while having a smaller top level. The energy transfer resistance should be in the range of the low trophic levels I and II. The low trophic level composed of detritus and primary producers (phytoplankton) was the main source of systemic energy.

The results suggested that the proportion of high nutrient levels in the system should be appropriately increased to make the ecosystem more mature and stable considering the current southwest Yellow Sea ecosystem. The total conversion efficiency of the system was 13.22%, and the energy conversion efficiency of each trophic level was higher than Lindeman's 1/10 conversion efficiency. However, the conversion efficiency between high trophic levels was generally higher than that between low trophic levels. The energy of the system's low trophic grades I and II was mostly in surplus and was not fully utilized with the filter-feeding organisms looking for detritus and plankton as the main food source. For example, the habitat of bivalve mollusks had caused a certain degree of influence, resulting in a reduction in the utilization of low trophic energy. The high conversion efficiency between high trophic levels might be due to the high-intensive fishing of economic fish species, such as hairtail, small yellow croaker, baby croaker, and pomfret, which all belong to high trophic levels.

Table 6 compares our results with those of other similar ecosystems, including the South Yellow Sea (Lin 2012; Lin et al. 2013), the East China Sea (Cheng et al. 2009), northern South China Sea (Cheng and Qiu 2010), the Gulf of California (Francisco et al. 2002), South Catalan Sea (Coll et al. 2006), and Gulf of Lions (BaNaru et al. 2013). The total primary production/total respiration (TPP/TR) of the southwest Yellow Sea ecosystem was 2.541, and the total primary production /total biomass (TPP/TB) was 50.326, indicating the ecosystem maturity was low. The value of TPP/TR with 2.541 was lower than that in the East China Sea (4.518) and the northern South China Sea (2.596), higher than that in the California gulf (1.038), the South Catalan Sea (1.180), the southern Yellow Sea (1.430), and the Gulf of Lions (2.090). The TPP/TB with value of 50.362 was lower than that in the East China Sea (91.047), higher than that in the southern Yellow Sea (41.270), the northern South China Sea (25.000), the California gulf (27.390), the Gulf of Lions

Table 6 Comparison of parameters of SYSE with those of the other sea areas

Parameter	This study	Southern Yellow Sea	East China Sea	Northern South China Sea	California gulf, Mexico	South Catalan Sea	Gulf of Lions
Total system throughput (t/km^2/a)	3997.260	4956.830	7211.000	13037.000	4224.000	1657.000	2995.000
Sum of all consumption (t/km^2/a)	1031.033	1954.970	1289.249	3324.185	2208.901	852.110	1480.100
Sum of all exports (t/km^2/a)	991.669		2518.649	3236.992	66.381	61.270	251.700
Sum of all respiratory flows (t/km^2/a)	643.731	1263.500	715.951	2027.590	1664.254	327.160	498.700
Sum of all flows into detritus (t/km^2/a)	1330.826	1194.210	2686.870	4448.650	284.116	416.910	764.600
Sum of all production (t/km^2/a)	1816.496	2108.160	3549.000	5895.000	2269.000	658.000	1572.800
Calculated total net primary production (t/km^2/a)	1635.400	1807.640	3234.600	5264.580	1728.190	386.680	1042.400
Net system production (t/km^2/a)	991.669	544.150	2518.649	3236.990		59.520	543.700
Total biomass (excluding detritus) (t/km^2/a)	32.473		35.527	212.761	63.095	58.990	68.900
Total primary production/total respiration	2.541	1.430	4.518	2.596	1.038	1.180	2.090
Total primary production/total biomass	50.362	41.270	91.047	25.000	27.390	6.550	15.100
Total biomass/total throughput	0.008		0.005	0.016	0.015	0.040	0.023
Mean trophic level of the catch	2.926		3.320	2.930	2.870	3.120	3.240
Connectance index	0.280	0.360		0.290	0.245	0.200	0.150
System omnivory index	0.217	0.210		0.239	0.332	0.190	0.210
Profit	0.612					0.670	0.670
Finn's cycling index	3.983			4.380		25.190	11.870
Finn's mean path length	2.444			2.476	3.500	4.270	3.990

(15.100), and the South Catalan Sea (6.550). The maturity of the southwest Yellow Sea ecosystem was higher than that of the East China Sea and lower than the southern Yellow Sea, the northern South China Sea, the California gulf, the South Catalan Sea, and the Gulf of Lions ecosystem, showing a lower maturity stage. Based on other relevant indicators judging the maturity of ecosystems, the value of SOI in SYSE was 0.28 and the connection index (CI) was 0.217, which was higher than that of the South Catalan Sea (0.200, 0.190) and the Gulf of Lions (0.150, 0.210), and lower than that of the northern South China Sea (0.290, 0.239). Compared with the southern Yellow Sea, the SOI values of the SYSE were lower and the CI values were higher. Compared with the California gulf, Mexico, the SOI values of the SYSE were higher and the CI values were lower. The values of Finn's cycling index (FCI) and Finn's mean path length (MPL) in SYSE were 3.983 and 2.444, respectively, which were similar to the northern South China Sea ecosystem value (4.380, 2.476), but much lower than the value of the South Catalan Sea and Gulf of Lions (25.190, 4.270; 11.870, 3.990).

SIA method provides a way to testify the credibility of results produced by Ecopath model, a mass-balanced model mostly depending on parameters input. Table 7 (Du et al. 2015; Lassalle et al. 2014; Kline and Pauly 1998; Nilsen

Table 7 R^2 of study areas between trophic levels estimated from SIA and Ecopath model

Study area	R^2	References
Southwest Yellow Sea	0.72	Present study
Xiamen Bay	0.70	Du et al. (2015)
Continental shelf of the Bay of Biscay	0.72	Lassalle et al. (2014)
Prince William Sound	0.97	Kline and Pauly et al. (1998)
Fjord in northern Norway	0.72	Nilsen et al. (2008)
South tropical lagoon	0.82	Milessi et al. (2010)
South Catalan Sea	0.48	Navarro et al. (2011)

et al. 2008; Milessi et al. 2010; Navarro et al. 2011) reviews studies examining the relationship estimated by these two independent methods, SIA and Ecopath. All R^2 values were greater than 0.70 indicating a high agreement between the two methods. This suggests that both the Ecopath model and SIA methods were proven by cross-validation to determine the trophic level of food webs. It also indicates that the information on the composition of the food used in Ecopath model was relatively accurate (Coll et al. 2006).

There was a clear linear correlation between results estimated from the Ecopath model compared with SIA in SYSE based on the isotope data, which means the modeling results in this study can be applied to other similar ecosystem simulations (Kline and Pauly 1998). This further underlines the importance of using both methods conjointly for a given ecosystem. Based on two key assumptions, SIA selected bivalve mollusks, filter-feeding organisms, as the benchmark; thus, the results were also affected by spatiotemporal sampling, such as pollution and other uncertainties. However, the Ecopath model was a static system mass balance model constructed by comprehensive average data. Decision in the design of functional group structures in the Ecopath model might be a source of uncertainty, although it was a trade-off result of complexity and simplicity.

In general, the above comparison with other similar ecosystem parameters showed that the functional groups of the SYSE were closely connected and the food network structure was relatively complex. Comparing to the current ecosystem, the results of SYSE Ecopath model correspond to what the SYSE is experiencing now. The SYSE is indeed still in a fragile, unstable, and immature stage with intensive exploitation from surrounding areas, such as over-fishery, coastal wetland loss, shrinking inputs of runoff and sedimentary, environmental pollution, etc.

This paper was the first study focusing on both the structure and energy flow of the SYSE, China, using the Ecopath model and SIA to testify its accuracy. However, due to the limited data collected from the field sampling survey, the accuracy and credibility of modeling results still need more verification. In addition, the Ecosim and Ecospace modules in the EwE model have been applied to specific regional-specific problems in the world. Time-driven variables, spatial variables, environmental variables, and management strategy variables are all important indicators which affected the function and structure of the SYSE. Further study of SYSE focusing on environmental variables changes based on Ecosim and Ecospace modules will present more detailed information on marine organism conservation of Yellow Sea Large Marine Ecosystem.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11356-021-16032-5.

Acknowledgements The authors are grateful to Jian Cheng in Zhejiang Ocean University; Hongmei Zhu in Xiamen University; and Prof. Cheng Huang in the School of Life Sciences, Nanjing University, for their valuable support in identifying specific species. Special thanks to Ms. Sarah Kennedy at Integrated Oceans Management, Fisheries and Oceans Canada, for her valuable suggestions and English rewording.

Author contribution Wei Wang contributed to analyze most of the data and wrote the initial draft of the paper. Junjie Wang carried out additional analyses. Ping Zuo contributed to the central idea, refining the ideas with instruction of SIA method and finalizing this paper. All authors read and approved the final manuscript.

Funding This work was supported by the National Key Research and Development Program of China under grant no. 2017YFC0506506 and no. 2017YFF0206905; the National Natural Science Foundation of China with grant no. 41473001; and Marine Science and Technology Innovation Project of Jiangsu Provincial Marine and Fisheries Bureau (HY2018-2).

Data and materials availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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